Case Study 2: SIOP Reporting

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Executive Summary

The objective of this case study was to architect a scalable Sales, Inventory & Operations Planning (SIOP) reporting solution. The primary challenge was transforming periodic forecast snapshots into a cohesive model that allows executives to compare predictive data (Resultant, Lag 0, Lag 1) directly against actual sales performance.

This dashboard was developed entirely within the Power BI Service cloud environment, demonstrating a cloud-native, platform-agnostic approach to enterprise data architecture. The resulting solution provides an intuitive, high-performance executive dashboard equipped with automated risk-flagging and dynamic metric filtering.

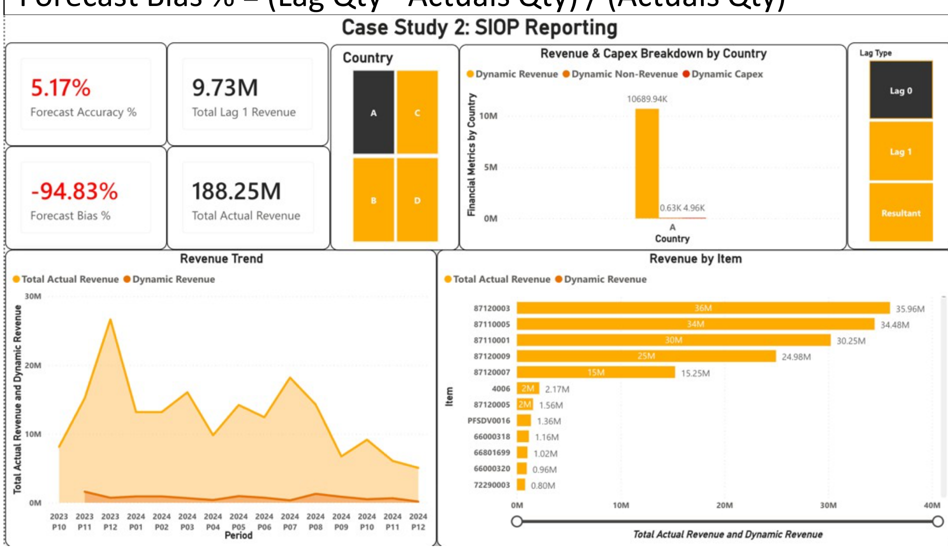
1. Data Handling & Architecture

To ensure a robust and performant solution, the data model was designed to handle continuous monthly updates without breaking downstream visualizations.

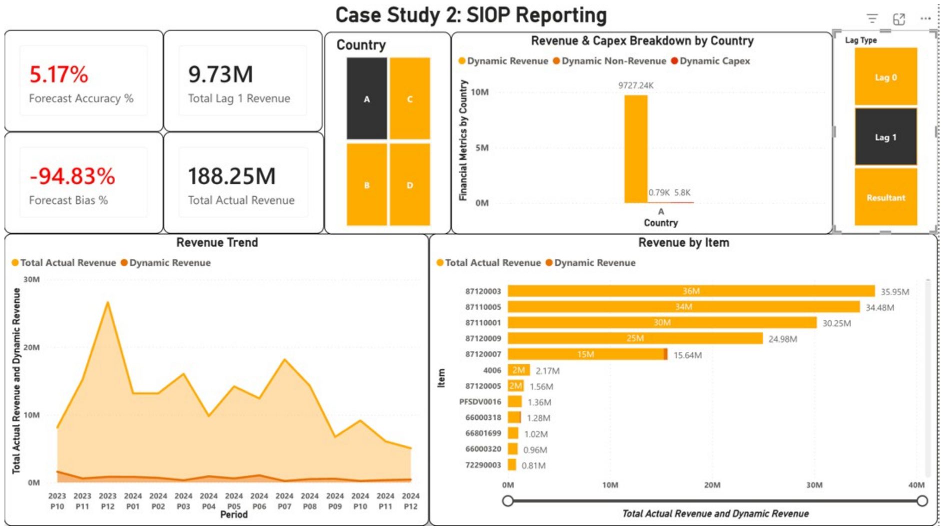
- **Data Consolidation & Merging:** The multiple forecast snapshots (AsofPeriods) were consolidated and mapped to their respective future forecast periods. The actual sales data was then merged with the transformed forecast data using a composite key logic based on Period, Item, and Country.
- **Cloud-Native Transformation:** To simulate a modern, decentralized enterprise workflow, all data modelling and visual filtering were executed directly via the Power BI Service environment. This approach ensures the dashboard is lightweight and does not rely on local desktop software dependencies for deployment.
- **Fault-Tolerant KPI Logic:** Key Performance Indicators (KPIs) were engineered using safe division practices.

Forecast Accuracy % = 1 - ABS(Lag Qty - Actuals Qty) / (Actuals Qty)

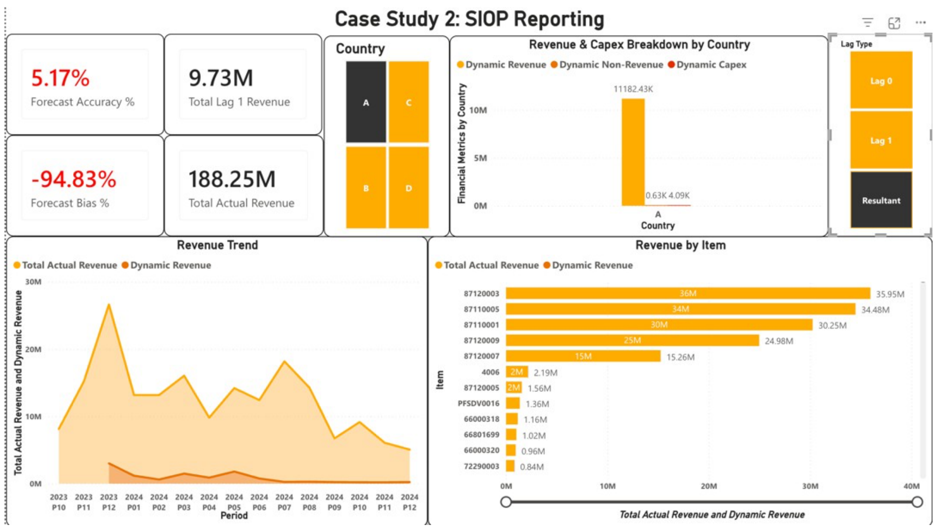
Forecast Bias % = (Lag Qty - Actuals Qty) / (Actuals Qty)



Multi-metric financial breakdown by Country Lag 0



Multi-metric financial breakdown by Country Lag 1

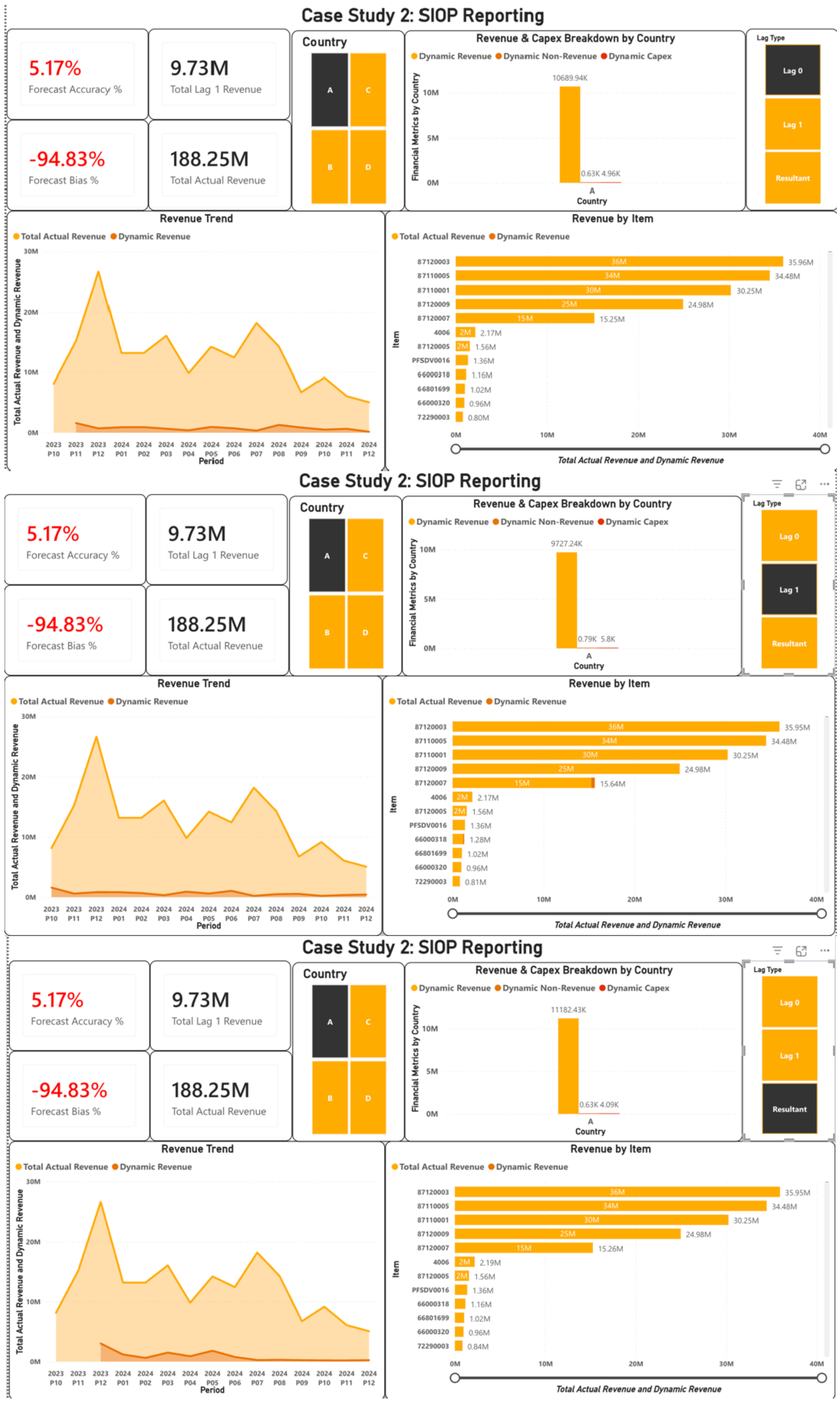


Multi-metric financial breakdown by Country Resultant

2. Dynamic Modelling & User Interaction

The prompt required a dynamic filtering mechanism to toggle the dashboard between Resultant, Lag 0, and Lag 1 snapshots. Rather than forcing the user to navigate separate report pages, a dynamic parameter architecture was implemented.

- **Dynamic Metric Toggling:** A disconnected parameter approach (using visual-level filtering and DAX SWITCH logic) was utilized to map user selections from the "Lag Type" slicer directly to the underlying snapshot columns.
- **Granular Drill-Down:** When an executive selects "Lag 0," the entire canvas including the Revenue Trend area chart and the multi-metric clustered bar chart instantly updates to reflect the Lag 0 data state, providing a seamless analytical experience.

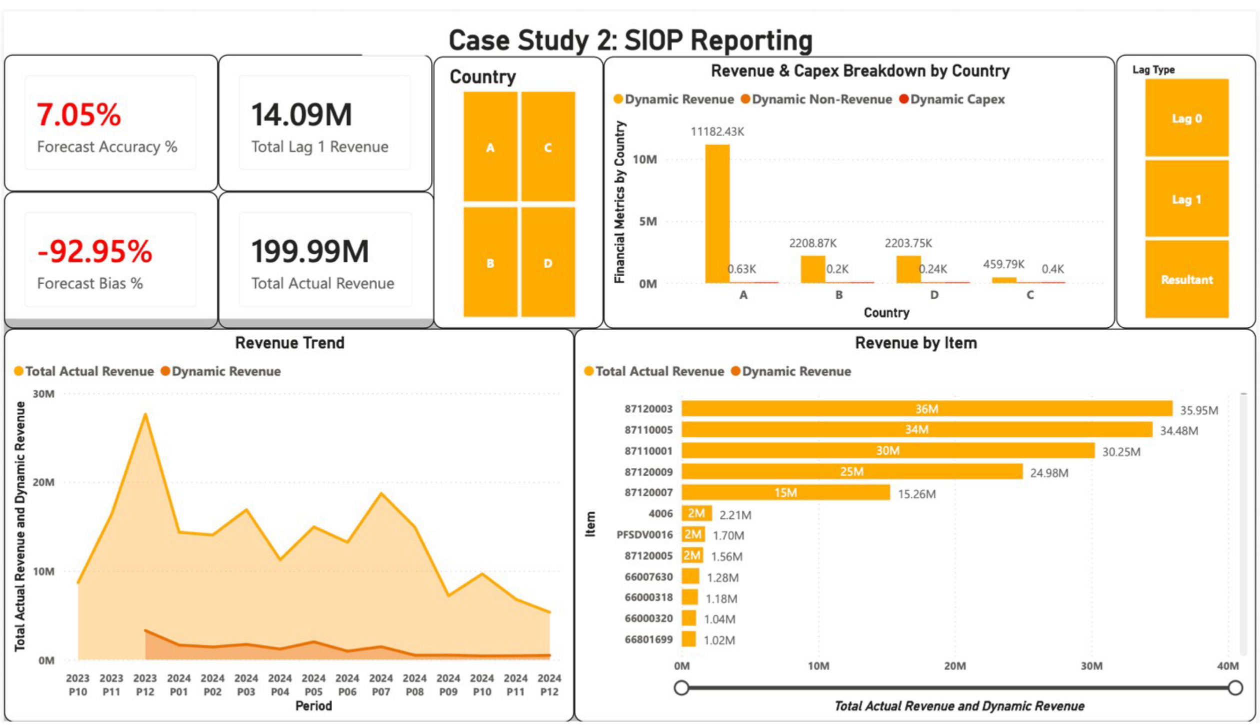


Visual-level filtering allows seamless toggling between forecast snapshot periods without cluttering the UI.

3. Dashboard Design & UX Philosophy

A dashboard is only as effective as its user adoption rate. The UI was intentionally designed to reduce cognitive load and highlight actionable business intelligence immediately.

- **Automated Risk Flagging (Conditional Formatting):** Executives do not have time to hunt for negative metrics. Conditional formatting logic was applied to the Forecast Bias % and Forecast Accuracy % KPIs. If the Bias falls into a negative threshold, the typography automatically shifts to red, instantly alerting leadership to underperforming forecasts.
- **Visual Hierarchy & Depth:** A high-contrast, modern enterprise colour palette was chosen. Visual borders and drop shadows were applied to the KPI cards to create a "floating" Z-index depth, mirroring modern web application design (SaaS) rather than a standard static spreadsheet.
- **Data Clarity:** Data labels were turned on for the bar charts, and clunky database column names were renamed to clear, plain-English titles (e.g., "Financial Metrics by Country") to ensure the dashboard is self-explanatory.



The final SIOP Executive Dashboard featuring conditional KPI formatting, intuitive layout, and a high-contrast enterprise aesthetic

4. **Predictive Model Strategy (Future Scalability)**

Addressing the Predictive Model Assessment Criteria

Currently, SIOP tracking relies on capturing manual monthly snapshots (the Lag system) to track how a forecast evolved over time. While this dashboard successfully visualizes that historical variance, the next evolutionary step for the company is to transition from *descriptive* analytics to *prescriptive* AI modelling.

Proposed Approach for Predictive Model Development: To fully optimize the SIOP pipeline, I propose developing a deep learning time-series model to automate and predict forecast variances before they occur.

a. **Model Selection:** Utilizing Recurrent Neural Networks (RNN), specifically Long Short-Term Memory (LSTM) architectures. LSTMs are uniquely suited for SIOP data because they can retain the "memory" of seasonal trends and previous lag variances over long sequences.
b. **Feature Engineering:** The model would be trained on historical inputs including Actuals Qty, historical Lag Qty variances, Item seasonality, and macroeconomic indicators.
c. **Enterprise Deployment:** By deploying this model on cloud infrastructure (such as AWS SageMaker or Oracle Cloud), the system could automatically generate the "Resultant" forecast based on historical Lag behaviours, drastically reducing manual forecast bias and optimizing inventory reserves.